1. Highest Relevance: Competency #6 — Identify and describe the mathematical aspects of a real-world problem situation, including identifying the significant variables

- Match: Students must identify key variables affecting payback time:
  - Annual fuel consumption (3.5M L)
  - Fuel saving rate (20%)
  - Diesel price (0.42 zeds/L)
  - Installation cost (2.5M zeds)
- Analysis: This is the foundational step of modeling—filtering mathematically relevant information from contextual text.
- ✅ Relevance: Very High

2. High Relevance: Competency #2 — Identify the key variables in a model

- Match: Closely related to #6, but specifically focuses on recognizing variables like "annual savings" and "total investment" as central to the model.
- Analysis: Understanding that "payback time = total cost ÷ annual savings" depends on correctly identifying these variables.
- ✅ Relevance: High

3. High Relevance: Competency #7 — Simplify or decompose a situation or problem to make it amenable to mathematical analysis

- Match: Real-world fuel savings depend on wind patterns, routes, speed, etc., but the problem assumes a fixed 20% reduction. Students must accept and apply this simplification.
- Analysis: Modeling inherently involves idealization. Although the simplification is provided, students must recognize its role in enabling calculation.
- ✅ Relevance: High

4. High Relevance: Competencies #10 / #11 — Use mathematical tools (variables, symbols, diagrams) to describe mathematical structures/relationships

- Match: Students express the relationship mathematically, e.g.:

$$\text{Years} = \frac{\text{Installation Cost}}{(\text{Annual Consumption} \times 0.20) \times \text{Price per Liter}} = \frac{2{,}500{,}000}{(3{,}500{,}000 \times 0.20) \times 0.42}$$

- Analysis: Even without formal algebraic notation, writing a clear arithmetic expression constitutes using mathematical tools to represent relationships.
- ✅ Relevance: High

5. Moderate Relevance: Competency #12 — Identify the constraints, assumptions, and simplifications in a mathematical model

- Match: Implicit assumptions include:
  - Constant annual fuel consumption
  - Stable 20% savings rate
  - Unchanging diesel price
  - No maintenance or operational costs for the kite sail
- Analysis: While the question doesn't require students to state these explicitly, awareness of them supports evaluating the reasonableness of the answer (e.g., "8.5 years under ideal conditions").
- ⚠️ Relevance: Moderate (present in modeling thinking, but not directly assessed)

6. Moderate Relevance: Competency #9 — Translate a problem into a standard mathematical representation or algorithm

- Match: The "payback period" concept is translated into a standard financial formula: total investment divided by annual return.
- Analysis: This reflects recognition of a canonical mathematical structure underlying a real-world decision problem.
- ✅ Relevance: Moderate

1. 最高度相关：第6项 — Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- 匹配点：学生必须识别出影响“回本时间”的关键变量：
  - 年油耗（3.5M L）
  - 节省比例（20%）
  - 柴油单价（0.42 zeds/L）
  - 初始投资（2.5M zeds）
- 分析：这是建模的第一步——从冗长文本中筛选出与目标（回本年限）相关的数学量。
- ✅ 相关性：极高

2. 高度相关：第2项 — Identify the key variables in a model

- 匹配点：与第6项紧密相关，但更强调“变量”本身（如年节省金额、总成本）。
- 分析：学生需意识到“回本时间 = 总成本 ÷ 年节省金额”，这依赖于对核心变量的识别。
- ✅ 相关性：高

3. 高度相关：第7项 — Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

- 匹配点：现实中的燃料节省可能受航速、风况、载重等影响，但题目已简化为“固定20%节省”。学生需接受这一简化，并忽略其他复杂因素。
- 分析：建模本质上是简化。本题虽由命题者完成部分简化，但学生仍需理解并应用这一理想化假设。
- ✅ 相关性：高

4. 高度相关：第10 / 11项 — Use mathematical tools (variables, symbols, diagrams) to describe mathematical structures/relationships

（注：第10和11项在原文中重复，视为一项）

- 匹配点：学生需用数学表达式建立关系，例如：

$$\text{Years} = \frac{\text{Installation Cost}}{\text{Annual Savings}} = \frac{2{,}500{,}000}{(3{,}500{,}000 \times 0.20) \times 0.42}$$

- 分析：虽然不强制使用代数符号，但清晰的算式本身就是“用数学工具描述关系”。
- ✅ 相关性：高

5. 中等相关：第12项 — Identify the constraints, assumptions, simplifications in a mathematical model

- 匹配点：隐含假设包括：
  - 每年航行条件相同（油耗恒定）
  - 20%节省率稳定
  - 柴油价格不变
  - 无维护成本
- 分析：题目未要求学生明确陈述这些假设，但理解它们有助于判断答案的合理性（如“约8.5年”是基于理想条件）。
- ⚠️ 相关性：中等（因题目未显性考查，但建模思维涉及）

6. 中等相关：第9项 — Translate a problem into a standard mathematical representation or algorithm

- 匹配点：将“回本时间”问题转化为“除法运算”（总成本 ÷ 年收益）。
- 分析：这是典型的“成本回收期”模型，属于标准财经数学表示。
- ✅ 相关性：中等